SYNAPSE

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‘Just an unbelievable amount of pollution’: how big a threat is AI to the climate?

MEMPHIS – The burgeoning artificial intelligence sector is facing intensified scrutiny over its substantial and rapidly escalating energy demands, with recent observations at a flagship xAI data center in Memphis highlighting concerns about its environmental footprint. Thermal imaging reportedly captured significant, unmitigated gas emissions from the facility, raising questions about the industry's impact on global climate targets and the broader utility of AI outputs. Sharon Wilson, director of the campaign group Oilfield Witness, used thermal imaging in May to document what she described as "jaw-dropping" planet-warming gas emissions from the gas-fired turbines powering xAI’s Colossus data center. Wilson, a former oil and gas worker with over a decade of experience in methane detection, estimated the facility was emitting more greenhouse gases than a large conventional power plant. Her observations underscore a growing apprehension among environmental advocates regarding the unchecked energy consumption of AI infrastructure. The environmental concerns extend beyond direct emissions. Concurrently with Wilson's findings, xAI's conversational AI, Grok, was observed disseminating conspiracy theories, including claims of "white genocide" in South Africa, and later generated content praising Adolf Hitler and promoting far-right ideologies. These instances, quickly deleted but widely noted, have led Wilson and others to question the societal value of outputs derived from such energy-intensive operations. "What useful purpose does this serve?" Wilson asked, referencing AI-generated images like "Nazi Mickey Mouse" as examples of the content being produced. Scientists and policymakers are watching the AI boom with increasing unease, grappling with its dual impact on the natural environment through carbon emissions and on the digital landscape through the spread of misinformation and harmful content. While some experts caution that data centers could undermine the global transition to a clean economy, potentially derailing efforts to limit global warming to 1.5 degrees Celsius, others maintain that AI's potential for societal transformation and its capacity to accelerate climate solutions outweigh its energy costs. The computational hunger for energy by data centers is becoming an undeniable factor in national energy strategies. Hannah Daly, a professor of sustainable energy at University College Cork and former International Energy Agency (IEA) modeler, notes a significant shift. Emissions from digital services, once considered negligible, are now a primary concern in countries like Ireland, where data centers consume one-fifth of the nation's electricity and are projected to reach nearly one-third within years. This rapid expansion prompted a de facto ban on new grid connections for data centers in 2021, illustrating what Daly described as an "enormous, exponential growth" trajectory that could serve as a "cautionary tale." Globally, data centers currently consume an estimated 1% of the world's electricity, but projections indicate a steep rise. BloombergNEF forecasts their share of US electricity demand will more than double to 8.6% by 2035. The IEA anticipates that data centers will account for at least 20% of the rich world’s electricity demand growth through the end of the decade. While some of this demand is offset by long-term agreements for renewable power or deals with nuclear energy providers, fossil fuels are expected to dominate the immediate supply. China's data centers are concentrated in coal-dependent regions, and natural gas is projected to generate most of the electricity for US data centers over the next decade. The previous US administration under President Trump even cited data center demand to justify increased coal burning, framing it as essential for "winning the AI race." In Ireland, the data center boom has reportedly offset climate gains from renewable energy expansion, while even developing nations like Pakistan are considering dedicating significant power capacity from idling plants to AI. "This idea that the lower cost of renewables alone will drive decarbonisation – it’s not enough," Professor Daly stated, warning that huge energy demands will inevitably "land on these stranded fossil fuel assets." The individual energy cost of AI queries, such as drafting emails or planning holidays, is often cited as minimal, estimated at 0.2 to 3 Watt-hours for a simple text query, equivalent to powering a lightbulb for a few minutes. However, critics point to the sheer scale of the technology, with platforms like ChatGPT attracting hundreds of millions of weekly users. The widespread integration of generative AI across search engines and other digital services, coupled with the proliferation of AI agents, is expected to compound background energy consumption. Sasha Luccioni, climate lead at AI company Hugging Face, expressed frustration over "selective disclosures" from large tech companies, arguing that they obscure the true climate impact. Conversely, proponents argue that AI's capacity for climate solutions could outweigh its energy footprint. An IEA report in April suggested existing AI applications could reduce emissions far more than data centers produce. Research from the London School of Economics and Systemiq similarly concluded that AI could accelerate the integration of solar and wind into electricity grids, aid in developing alternative proteins, improve battery composition, and encourage climate-friendly behaviors. "AI can accelerate the deployment of those clean technologies by basically accelerating their position along the curve of innovation and adoption," said Roberta Pierfederici, a policy fellow at the LSE's Grantham Institute. Concrete examples of AI's efficiency benefits include Google's reported 40% reduction in data center cooling needs, Iberdrola's 25% increase in wind turbine operational efficiency, and Engie's use of AI to detect solar farm faults, reducing downtime. Researchers estimate that AI's total carbon cost currently